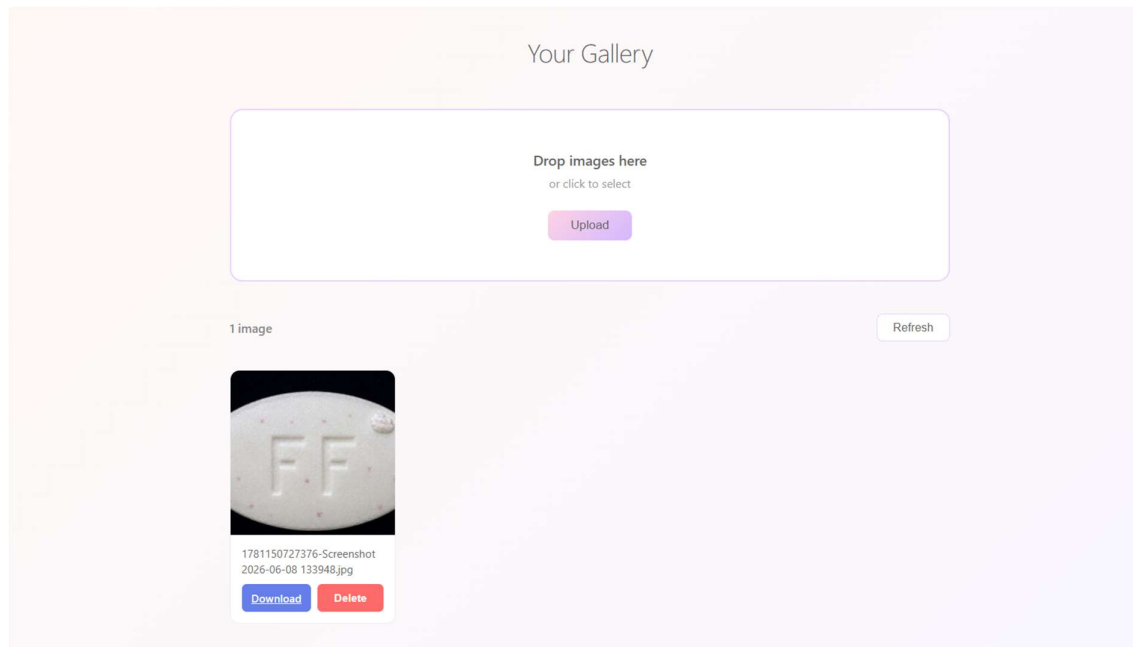


Day-4: Power BI Quality Dashboard and Frontend Integration

Frontend Interface:

The frontend was integrated using HTML, CSS, Javascript and a interface was created where the user can upload the file, view it, delete it, download it if needed. Then that file gets saved into a watcher folder locally on the computer.



Power BI:

The inference logged by python is converted to a csv file which is then read by Power BI and it is displayed in the dashboard.

The dashboard consists of :

- 1]Model Metrics (mAP, Precision, Recall, F1-score).
- 2]Present image prediction class, confidence score.
- 3]Total number of pills inspected.
- 4]Defect rate.
- 5]A pie chart displaying the quality distribution.
- 6] Bar chart displaying number of defects per batch.
- 7]Batch wise quality comparison.
- 8]Defect wise Heatmaps.

AI-Powered Tablet Visual Inspection System Dashboard

Precision

92.70%

Recall

88.80%

map

94.00%

F1

90.70%

colour

Predicted Class

65.16%

Confidence Score

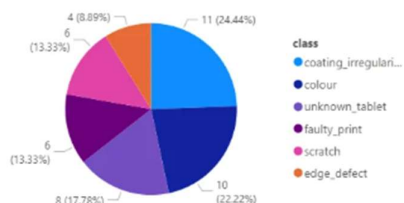
34

Total No.Of Pills Inspected

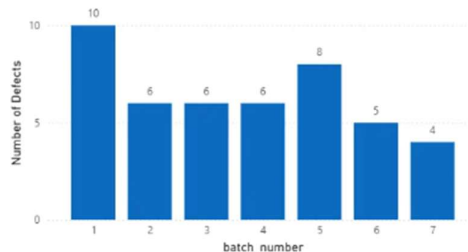
100.00%

Overall Defect Rate

Quality Distribution



Defects Per Batch Graph



Batch-Wise Quality Comparison



Defect Type Heatmap

batch_number	coating_irregularities	colour	edge_defect	faulty_print	scratch	unknown_tablet	Total
1	4			2		4	10
2	1				3	2	6
3	1		2		3		6
4	2	3	1				6
5		5		1		2	8
6	1	1	1	2			5
7	2	1		1			4
Total	11	10	4	6	6	8	45